

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Scenario-Based : 20%

QUESTIONS:

Total Marks: 50

Annexure A

Code of Conduct:

I Siva kumar M(192521057)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Sivakumar m

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

CO-2 -Analytical Problem Solving

Q1. RSA Encryption Analysis

Given $p=11$, $q=13$

- Compute n and $\phi(n)$
- Choose $e=7$ and find the private key d
- Encrypt the message $M=9$
- Decrypt the ciphertext and verify the result
- Analyze how changing e affects security

Ans Given

- $p = 11$
- $q = 13$
- $e = 7$
- Message $M = 9$

(a) Compute n and $\phi(n)$

Step 1: Compute n

$$n = p \times q = 11 \times 13 = 143$$

Step 2: Compute Euler's Totient

$$\begin{aligned}\phi(n) &= (p - 1)(q - 1) \\ &= (11 - 1)(13 - 1) \\ &= 10 \times 12 = 120\end{aligned}$$

Answer:

- $n = 143$
 - $\phi(n) = 120$
-

(b) Choose $e = 7$ and find the private key d

The private key satisfies

$$d \times e \equiv 1 \pmod{\phi(n)}$$

So,

$$7d \equiv 1 \pmod{120}$$

Find the multiplicative inverse of 7 modulo 120.

Try:

$$\begin{aligned} 7 \times 103 &= 721 \\ 721 \div 120 &= 6 \text{ remainder } 1 \end{aligned}$$

Hence,

$$721 \equiv 1 \pmod{120}$$

Therefore,

$$\boxed{d = 103}$$

Keys:

- Public Key = (7, 143)
 - Private Key = (103, 143)
-

(c) Encrypt the message $M = 9$

Encryption formula:

$$C = M^e n$$
$$C = 9^7 143$$

Using repeated squaring:

$$9^2 = 81$$
$$9^4 = 81^2 = 6561$$
$$6561 143 = 126$$

Now,

$$9^7 = 9^4 \times 9^2 \times 9$$
$$= 126 \times 81 \times 9$$

First,

$$126 \times 81 = 10206$$
$$10206 143 = 53$$

Next,

$$53 \times 9 = 477$$
$$477 143 = 48$$

Therefore,

$$C=48$$

(d) Decrypt the ciphertext

Ciphertext:

$$C = 48$$

Decryption formula:

$$M = C^d n$$
$$M = 48^{103} 143$$

Using modular exponentiation,

$$48^{103} 143 = 9$$

Hence,

$$M=9$$

The decrypted message is the same as the original message.

Verification:

Original Message:

$$M = 9$$

Decrypted Message:

$$M = 9$$

✓ Therefore, RSA encryption and decryption are verified.

(e) Analyze how changing e affects security

- e must satisfy:
 - $1 < e < \phi(n)$
 - $\gcd(e, \phi(n)) = 1$
- A small value of e (such as 3 or 5) makes encryption faster but can introduce vulnerabilities if padding is not used.
- A larger value of e increases encryption time but may reduce some risks associated with very small exponents.
- The commonly used value is: $e = 65537$

Q2 Block Cipher Mode Evaluation

Given plaintext blocks:

$P=[1010,1010,1111,1010]$,

Key = $K=1100$

- Encrypt using ECB and CBC modes (Assume IV = 0000)
- Compare the ciphertext outputs
- Analyze which mode hides patterns better and explain why

Ans

Given

- Plaintext blocks:
 - $P_1 = 1010$
 - $P_2 = 1010$
 - $P_3 = 1111$
 - $P_4 = 1010$
- Key:
 - $K = 1100$
- IV (Initialization Vector):
 - 0000

Assumption: Encryption is performed using XOR.

$$C = P \oplus K$$

(a) Encrypt using ECB Mode

Formula

$$C_i = P_i \oplus K$$

Block 1

$$\begin{aligned} &1010 \\ &\oplus 1100 \\ &= 0110 \\ &C_1 = 0110 \end{aligned}$$

Block 2

Same plaintext.

$$\begin{aligned} &1010 \oplus 1100 = 0110 \\ &C_2 = 0110 \end{aligned}$$

Block 3

$$\begin{aligned} &1111 \oplus 1100 = 0011 \\ &C_3 = 0011 \end{aligned}$$

Block 4

$$\begin{aligned} &1010 \oplus 1100 = 0110 \\ &C_4 = 0110 \end{aligned}$$

ECB Ciphertext

[0110, 0110, 0011 0110]

Encrypt using CBC Mode

Formula

First block:

$$C_1 = (P_1 \oplus IV) \oplus K$$

Remaining blocks:

$$C_i = (P_i \oplus C_{i-1}) \oplus K$$

Block 1

$$\begin{aligned} P_1 \oplus IV \\ 1010 \oplus 0000 &= 1010 \end{aligned}$$

Now,

$$\begin{aligned} 1010 \oplus 1100 &= 0110 \\ C_1 &= 0110 \end{aligned}$$

Block 2

$$\begin{aligned} 1010 \oplus 0110 &= 1100 \\ 1100 \oplus 1100 &= 0000 \\ C_2 &= 0000 \end{aligned}$$

Block 3

$$1111 \oplus 0000 = 1111$$

$$1111 \oplus 1100 = 0011$$

$$C_3 = 0011$$

Block 4

$$1010 \oplus 0011 = 1001$$

$$1001 \oplus 1100 = 0101$$

$$C_4 = 0101$$

CBC Ciphertext

[0110, 0000, 0011 0101]

(b) Compare Ciphertexts

Block	ECB	CBC
1	0110	0110
2	0110	0000
3	0011	0011
4	0110	0101

(c) Which mode hides patterns better?

ECB

- Same plaintext → Same ciphertext.
- Patterns remain visible.
- Less secure.

CBC

- Every block depends on the previous ciphertext.
- Identical plaintext blocks produce different ciphertexts.
- Better security.

Conclusion:

CBC mode hides patterns better because every block depends on the previous ciphertext block and the Initialization Vector (IV).

Q3 DES vs 3-DES vs Blowfish Performance Study

Given:

- a. DES key size = 56 bits, time = 2 ms/block**
- b. 3-DES key size = 168 bits, time = 6 ms/block**
- c. Blowfish key size = 128 bits, time = 1 ms/block**

For 1000 blocks of data:

I .Calculate total encryption time for each algorithm

II .Analyze trade-offs between security and performance

III .Recommend the best algorithm for secure real-time communication

Ans

Given

Algorithm	Key Size	Time per Block
DES	56 bits	2 ms
3-DES	168 bits	6 ms
Blowfish	128 bits	1 ms

Number of blocks = 1000

(i) Calculate Total Encryption Time

DES

$$1000 \times 2 = 2000 \text{ ms} = 2 \text{ seconds}$$

3-DES

$$1000 \times 6 = 6000 \text{ ms} = 6 \text{ seconds}$$

Blowfish

$$1000 \times 1 = 1000 \text{ ms} = 1 \text{ second}$$

Algorithm	Total Encryption Time
DES	2000 ms (2 s)
3-DES	6000 ms (6 s)
Blowfish	1000 ms (1 s)

(ii) Analyze Trade-offs

DES

- Small key size (56 bits)
- Fast encryption
- Weak security

3-DES

- Large key size (168 bits)
- Very secure
- Slowest algorithm

Blowfish

- 128-bit key
 - Strong security
 - Fast encryption
 - Efficient memory usage
-

(iii) Recommendation

Blowfish is the best choice for secure real-time communication because it provides strong security with the fastest encryption time.

Q4 Diffie-Hellman Key Exchange Analysis

Given:

- a. $p=23$, $g=5$**
- b. User A private key = 6**
- c. User B private key = 15**
- d. Compute public keys for A and B**
- e. Compute the shared secret key**
- f. Demonstrate how a Man-in-the-Middle attacker could alter the exchange**
- g. Suggest a secure method to prevent the attack**

Ans

(a) Given Prime Number

$$p = 23$$

(b) Given Generator

$$g = 5$$

(c) Given Private Keys

User A private key:

$$a = 6$$

User B private key:

$$b = 15$$

(d) Compute Public Keys

Formula

$$\text{Public Key} = g^{\text{Private Key}} \pmod{p}$$

User A

$$\begin{aligned} A &= 5^6 \pmod{23} \\ 5^2 &= 25 \equiv 2 \pmod{23} \\ 5^4 &= 2^2 = 4 \\ 5^6 &= 5^4 \times 5^2 \\ &= 4 \times 2 = 8 \end{aligned}$$

Public Key of A = 8

User B

$$\begin{aligned} B &= 5^{15} \pmod{23} \\ 5^1 &= 5 \\ 5^2 &= 25 \equiv 2 \pmod{23} \\ 5^4 &= 2^2 = 4 \\ 5^8 &= 4^2 = 16 \\ 5^{15} &= 5^8 \times 5^4 \times 5^2 \times 5^1 \\ &= 16 \times 4 \times 2 \times 5 \\ 16 \times 4 &= 64 \equiv 18 \pmod{23} \\ 18 \times 2 &= 36 \equiv 13 \pmod{23} \\ 13 \times 5 &= 65 \equiv 19 \pmod{23} \end{aligned}$$

Public Key of B = 19

(e) Compute the Shared Secret Key

User A

$$\begin{aligned}K &= 19^6 \text{ } 23 \\19^2 &= 361 \equiv 16 \\19^4 &= 16^2 = 256 \equiv 3 \\19^6 &= 19^4 \times 19^2 \\&= 3 \times 16 = 48 \\48 &\equiv 2 \pmod{23} \\K &= 2\end{aligned}$$

Verification by User B

$$\begin{aligned}K &= 8^{15} \text{ } 23 \\8^2 &= 64 \equiv 18 \\8^4 &= 18^2 = 324 \equiv 2 \\8^8 &= 2^2 = 4 \\8^{15} &= 8^8 \times 8^4 \times 8^2 \times 8 \\&= 4 \times 2 \times 18 \times 8 \\4 \times 2 &= 8 \\8 \times 18 &= 144 \equiv 6 \\6 \times 8 &= 48 \equiv 2 \pmod{23} \\K &= 2\end{aligned}$$

Shared Secret Key = 2

(f) Demonstrate a Man-in-the-Middle (MITM) Attack

1. User A sends public key 8 to User B.
2. The attacker intercepts the key and sends a fake public key to User B.
3. User B sends public key 19 to User A.

4. The attacker intercepts it and sends another fake public key to User A.
 5. User A and User B unknowingly establish different secret keys with the attacker.
 6. The attacker can decrypt, modify, and re-encrypt all messages without either user knowing.
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(g) Suggest a Secure Method to Prevent the Attack

The attack can be prevented by using:

- Digital Signatures
- Digital Certificates
- Public Key Infrastructure (PKI)
- Authenticated Diffie–Hellman Protocol

Digital signatures and certificates authenticate the exchanged public keys, preventing attackers from replacing them during the key exchange.

Q5 ECC vs RSA Comparison

Given:

- a. RSA key size = 1024 bits**
- b. ECC key size = 160 bits (equivalent security)**
- c. Device processing capacity = limited (IoT device)**
- d. Analyze computational cost for both systems**
- e. Estimate which consumes less power and memory**
- f. Justify which cryptosystem is better for the given scenario with reasoning**

Ans

(a) Given

- RSA key size = 1024 bits
-

(b) Given

- ECC key size = 160 bits (provides security approximately equivalent to RSA 1024-bit)
-

(c) Given

- Device processing capacity = Limited (IoT Device)
-

(d) Analyze Computational Cost

RSA

- Uses a 1024-bit key, so computations are more complex.
- Requires more CPU processing time.
- Encryption and decryption are slower.
- Higher computational cost.

ECC

- Uses a much smaller 160-bit key for the same security level.
- Requires fewer mathematical operations.
- Faster execution.
- Lower computational cost.

Comparison: ECC has a lower computational cost than RSA.

(e) Estimate Which Consumes Less Power and Memory

RSA

- Larger key size requires more memory.
- Uses more processor time.
- Consumes more battery power.

ECC

- Smaller key size requires less memory.
- Uses less processor time.
- Consumes less battery power.

Comparison: ECC consumes less power and memory than RSA.

(f) Justify Which Cryptosystem is Better

For an IoT device with limited processing capacity, ECC is the better choice because:

- It provides the same security as RSA with a much smaller key size.
- It has lower computational cost.
- It requires less memory.
- It consumes less power.
- It performs faster, making it suitable for battery-powered and resource-constrained devices.

Conclusion:

ECC is recommended for the given IoT scenario because it is more efficient while maintaining equivalent security.